

```
/* Show all widgets that are contained by the window */
gtk_widget_show_all(GTK_WIDGET(window));

/* Start the main event loop. */
g_print("main: calling gtk_main()\n");
gtk_main();

/* Display a message to the standard output and exit. */
g_print("main: returned from gtk_main and exiting with success!\n");
return EXIT_SUCCESS;
}
```

Save the file, say *sampleGUI.c*. Now all you have to do is compile the program. Remember that you will need GTK+ on your system to be able to compile the program successfully; otherwise, it will not work. There are some specific flags that need to be passed while compiling. The flags required can be known using the *pkg-config* tool. *pkg-config --list-all* | sort will give you the flags. Issue the following command to compile the program for now:

```
[aditya@archie: ~] > gcc -Wall -g sampleGUI.c \
'pkg-config --cflags gtk+-2.0' -o sampleGUI
```

The detailed information about *pkg-config* help is available in the documentation of Maemo and is not a part of the discussion here. The *pkg-config* automation is possible, but, at this juncture, it is better to use manual flags. If some problems still persist, you might be missing some libraries that need to be explicitly linked. This can be sorted out using the *pkg-config* libraries:

```
[sbox-FREMANTLE_X86: ~] > gcc -Wall -g sampleGUI.c \
'pkg-config --cflags gtk+-2.0' -o sampleGUI \
'pkg-config --libs gtk+-2.0'
```

If you still encounter any compilation errors, the best way is to find someone on IRC channels and work these out. If all goes well, you can run the application using the following command:

```
[aditya@archie: ~] > ./sampleGUI
```

It will not work unless you have started the virtual X server. Now, to run the application on the actual environment emulator, run *xephyr* as:

```
[aditya@archie: ~] > Xephyr 2-host-cursor -screen 800x480x16 -dpi 96 -ac -kb &
```

Figure 1 shows the emulator start up. It should be outside the scratchbox environment. Back inside the scratchbox, give it explicit instructions to use the local socket and run GUI applications under the virtual server:

```
[aditya@archie: ~] > export DISPLAY=:2
[aditya@archie: ~] > ./gtk_helloworld-1
main: calling gtk_main
```

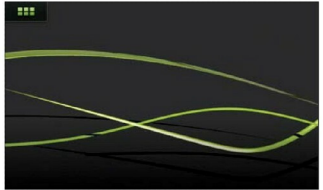


Figure 1: Emulator startup window

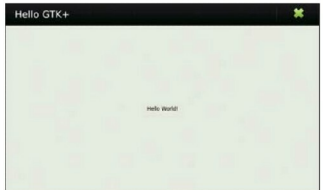


Figure 2: The 'Hello World!' dummy application

End the process with *Ctrl+c*. Now start the application framework from *af-sb-init.sh* start. It's a script to run the emulator environment. The screen similar to the one shown in Figure 2 will be presented to you.

## What's good about Maemo

The best thing about Maemo is its open source software. More importantly, the entire documentation has been released under the Creative Commons license (you can find it on [www.maemo.org](http://www.maemo.org)).

In addition, the community has developed support for the Qt framework and it's also possible to develop applications in languages other than C, such as C++ or Python. With the added support to create a modular structure for the widgets, Maemo development of late has been wonderfully easy and very intuitive. And with more community involvement, it could perhaps evolve into the best mobile computing platform in the world. 

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National Talent Scholar, Aditya Shevade, is a final year electronics engineering student, who takes keen interest in programming and electronic design. A Linux user for more than three years, he enjoys playing the keyboard and is a good photographer. To know more about him, visit [www.adityashevade.com](http://www.adityashevade.com)